

STRATA: A Stratified Process-Mining Framework for HVAC Fault Detection and Diagnosis

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Abstract

Faults in building HVAC equipment persist because the tools meant to catch them either raise alarms that operators learn to ignore, or flag anomalies that nobody can act on. STRATA turns ordinary one-minute trend data into an event log along the control sequence, discovers process models at the device and unit level, and scores each day through eight calibrated, physically interpretable channels against a measured noise floor, so that every alarm names a location, states its evidence and is held to a false-alarm budget. It was built under a falsification protocol: every hypothesis was pre-registered with the observation that would refute it, every result was recomputed independently before it was quoted, and automated tests pin the published numbers and the falsifiers that fired.

On five public LBNL benchmark systems, including the two fan-powered terminal units for which no detection results had been published before, STRATA detects 158 of 175 seeded faults, most on the first day, at 26 false-alarm days in 480 held-out fault-free days (1 to 9.4% per system). Two systems were onboarded blind, by configuration alone. The alarm names the correct terminal unit in 37 of the 50 detections with known ground truth and a wrong one in none, and the faulted subsystem in 83% of all detections. The false-alarm budget holds under injected sensor error, change-of-value logging and schedule shifts (worst 7 of 96 days) and on one measured rooftop unit (3 of 49).

Process mining helps to get detection but is not detecting on its own. The strata and the event abstraction carry the localisation, and invariants that discovery surfaced became several of the rules that detect. Conformance checking of the discovered models, pre-registered as the detector, was refuted: over 199 scenarios on six systems, no fault is caught by the models alone. A principal-component baseline detects three fewer of the 73 scenarios it shares and cannot name the device. A third-party Guideline 36 rule battery, held to the same gate, matches STRATA within one detection on the air handler and detects a third as many faults on the terminal units, so the pre-registered improvement claim over current practice is withdrawn there. The evaluation also produced eight machine-verified errata in the benchmark datasets. Every number regenerates from the public repository.

1 Introduction

Buildings and construction account for 32% of global final energy use and 34% of energy-related carbon dioxide emissions [20]. HVAC accounts for up to half of an individual building’s consumption [3] and for about 30% of all commercial building energy in the United States [8], and 15–30% of that is wasted through faulty operation and poor control [3,11]; correcting the common faults saves 5–30% of whole-building energy [3,17]. The faults persist even where detection is installed. Across

60,000 pieces of equipment in 317 commercial buildings, Crowe and colleagues found 245 faults per building per month and 40% of air handlers in a faulted state on any given day, in buildings that run modern automation systems with alarms [6]. Seven of their ten most common faults are behaviour-based: wrong schedules, unreleased overrides, missed setpoint resets, broken economiser sequences. A threshold alarm or a pre-written rule [18] catches such a fault only if someone wrote that rule in advance [10]. The public benchmarks used in this paper seed component, sensor and control-loop faults and none of that class, so this argument motivates the design and is not tested by it.

Research has gone the other way. Machine learning has been surveyed across 211 HVAC fault-detection studies [3], with accuracies saturated at 97–99% on the standard benchmarks [16, 19] and interpretability added afterwards at the level of input features [5, 7, 12, 22, 23], which tells an operator that a sensor mattered, not what the building should have been doing. Of 65 primary studies audited in 2025, two shared code links, one of them broken, and 72% do not say whether their data is public [13].

What an operator needs is a detector that works from the trend data already recorded, names the unit and subsystem, explains the alarm in the physics of the equipment, is held to a false-alarm rate measured on the building’s own fault-free days, and reaches a new building by configuration rather than new code. Process mining, the one methodology built to model the sequence in which a system moves between states, has not been brought to this problem: to our knowledge no published work applies event-log process mining to building HVAC fault detection. The 211-study survey contains no process-mining entry [3], the reviews of process mining in Industry 4.0 and on sensor data contain no building HVAC application [1, 4], the Lawrence Berkeley National Laboratory taxonomy of fault-detection methods has no such leaf [9], and our own exact-phrase searches of the OpenAlex corpus return none [15].

STRATA applies process mining in stratified form: sensor streams are abstracted into events along ASHRAE Guideline 36 control sequences [2], Petri nets are discovered per device and per unit [14, 21], and the channels that read the log sit beside physics rules and residuals that read the raw signals, each calibrated on its own noise. Our first hypothesis, that the discovered net would itself be the detector, was pre-registered and refuted: a single hand-written rule out-covered the model channel where it was strongest. What settled is the division of labour. The strata and the event abstraction give every alarm its location, invariants that discovery surfaced became several of the rules that detect, and conformance checking of the discovered nets contributes no detection of its own. Process mining helps to get detection but is not detecting on its own.

The paper contributes (1) the framework and the detector it yields, evaluated on five public benchmark systems including the two fan-powered terminal units for which no detection results had been published before; (2) a settled, pre-registered answer to the process-mining question that motivated the work; (3) an evaluation protocol of pre-registration, honoured falsifiers, independent recomputation and full artefact regeneration; and (4) eight machine-verified errata in the benchmark datasets.

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